Offline TA Session

EXAMPLE 3.7.A / PROBLEM 1

Question 1. An insurance company believes that people can be divided into two classes — those that are accident prone and those that are not. Their statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a non-accident-prone person. If we assume that 30 percent of the population is accident prone, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?

Solution:

Probability of being accident-prone (A) : $P(A) = 0.30$

Probability of being non-accident-prone (A^c) : $P(A^c) = 1 - 0.30 = 0.70$

Probability of an accident given someone is accident-prone: $P(Accident \mid A) = 0.40$

Probability of an accident given someone is not accident-prone: $P(Accident \mid A^c) = 0.20$

$P(Accident) = P(Accident \mid A) \cdot P(A) + P(Accident \mid A^c) \cdot P(A^c)$
$P(Accident) = (0.40 \cdot 0.30) + (0.20 \cdot 0.70)$
$P(Accident) = 0.12 + 0.14$
$P(Accident) = 0.26$

EXAMPLE 3.7.E / PROBLEM 2

Question 2. A laboratory blood test is 99 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability a person has the disease given that his test result is positive?

Solution:

0.5% of the population has the disease: $P(D) = 0.005$

Rest of the healthy population: $P(D^c) = 1 - 0.005 = 0.995$

True Positive rate (Sensitivity): i.e., the test is 99% effective when the disease is present = $P(+ve \,|\, D) = 0.99$

False Positive rate: The test yields a positive result for 1% of healthy people = $P(+ve \,|\, D^c) = 0.01$

$P(+ve) = P(+ve \,|\, D) \,.\, P(D) + P(+ve \,|\, D^c) \,.\, P(D^c)$ = (0.99)(0.005) + (0.01)(0.995) = 0.0149

$$P(D \,|\, +ve) = \frac{P(+ve \,|\, D)P(D)}{P(+ve)} = \frac{(0.99)(0.005)}{0.01490} = \frac{0.00495}{0.01490} = 0.3322$$

EXAMPLE 4.5.H / PROBLEM 3

Question 3: Suppose there are 20 different types of coupons and suppose that each time one obtains a coupon it is equally likely to be any one of the types. Compute the expected number of different types that are contained in a set for 10 coupons.

Solution:

$$X_i = \begin{cases} 1 & \textbf{if coupon type } i \textbf{ is contained in the set} \\ 0 & \textbf{otherwise} \end{cases}$$

$$X = \sum_{i=1}^{20} X_i$$

$$E[X] = E\left[\sum_{i=1}^{20} X_i\right] = \sum_{i=1}^{20} E[X_i]$$

Probability that a specific coupon type is chosen at least once across 10 trials

$$P(X_i = 1) = 1 - P(X_i = 0) = 1 - \left(\frac{19}{20}\right)^{10}$$

The final calculated expected value

$$E[X] = 20 \times \left[1 - \left(\frac{19}{20}\right)^{10}\right] \approx 8.0253$$

EXAMPLE 4.9.A / PROBLEM 4

Question 4: Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.
(a) What can be said about the probability that this week's production will exceed 75?
(b) If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?

Solution:

A). Because X is non-negative random variable and we only know its mean, we use Markov's Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$P(X \geq a) \leq P(X \geq 75) \leq \frac{E[X]}{75} = \frac{50}{75} = \frac{2}{3}$$

B). Since we know variance and Mean, applying Chebyshev's inequality

$Var(X) = 25$
$\sigma = \sqrt{25} = 5$

If the problem gives you a specific range (like "between 40 and 60") alongside the mean (μ) and the standard deviation (σ), you can solve for k using either the upper or lower boundary

$$distance = k\sigma = \frac{Boundary - Mean}{StandardDeviation} = \frac{(40{,}60) - 50}{5} = \pm|10|$$

$$k = \frac{10}{5} = 2$$

$$P(|X - \mu| \geq k\sigma \leq \frac{1}{k^2} = \frac{1}{4}$$

$$P(|X - 50| \geq 10 \leq \frac{1}{2^2} = \frac{1}{4}$$

But this range is outside the interval and we are looking for the probability of falling inside the interval.

$$P(40 < X < 60) = 1 - P(|X - 50| \geq 10) \geq 1 - \frac{1}{4} = \frac{3}{4}$$

The probability that this week's production will be between 40 and 60 is at least $\frac{3}{4}$ or 0.75

EXAMPLE CHAPTER 4 UNSOLVED QUESTION 4 / PROBLEM 5

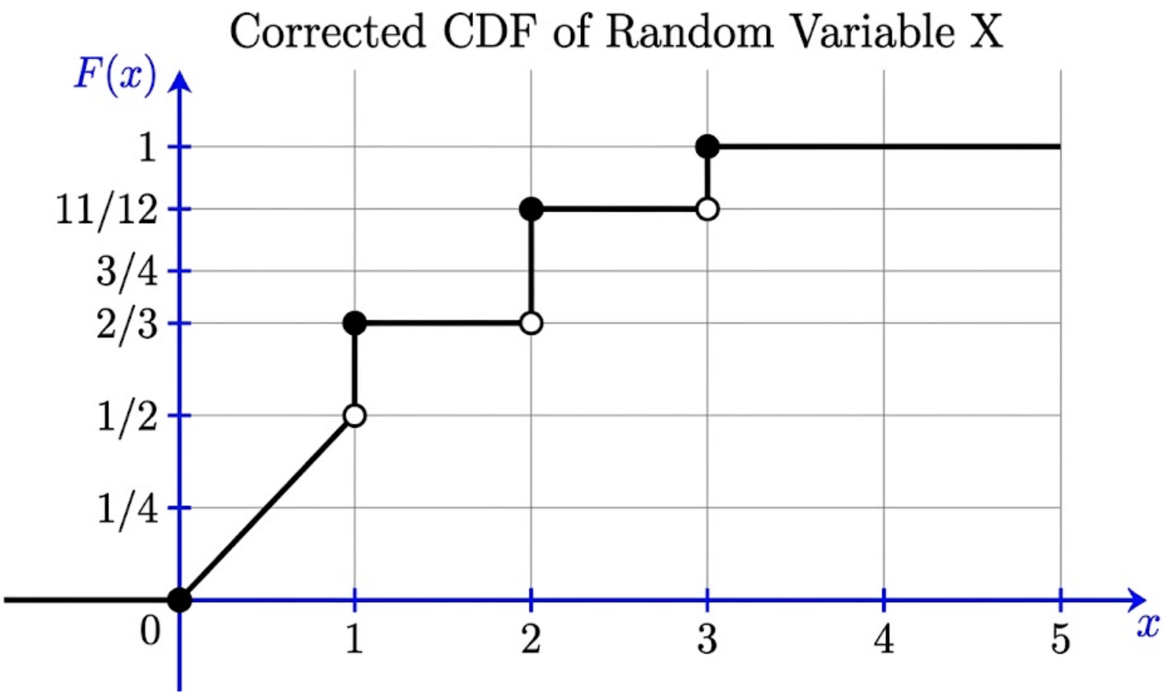
Question 5: The Distribution function of the random variable X is given

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 1 \\ \frac{2}{3} & 1 \leq x < 2 \\ \frac{11}{12} & 2 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

a. Plot this distribution function
b. What is $P(X) > \frac{1}{2}$
c. What is $P(2 < X \leq 4)$
d. What is $P(X < 3)$
e. What is $P(X = 1)$

Solution:

The above is the CDF function and not PDF function



B. $P\left(X > \frac{1}{2}\right) = 1 - F\left(\frac{1}{2}\right) = 1 - \frac{\frac{1}{2}}{2} = 1 - \frac{1}{4} = \frac{3}{4}$

C. $P(2 < X \leq 4) = F(4) - F(2) = 1 - \frac{11}{12} = \frac{1}{12}$

D. $P(X < 3) = F(3^{-}) = \frac{11}{12}$

E. $P(X = 1) = F(1) - F(1^{-}) = \frac{2}{3} - \frac{1}{2} = \frac{4}{6} - \frac{3}{6} = \frac{1}{6}$